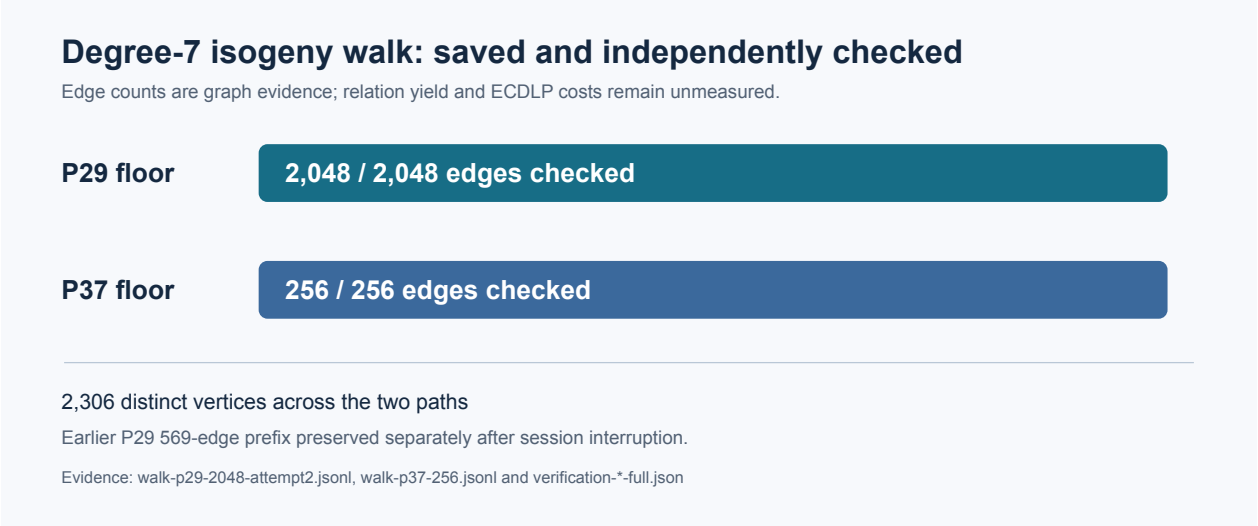
1. Degree-7 isogeny walk search: verified graph, pending relation study

Research record, 16 October 2026. This report covers graph enumeration only. The two requested finite paths contain 2,304 directed degree-7 edges and 2,306 distinct vertices. Three-summand relation yield, useful rank and end-to-end ECDLP time have not yet been measured on these paths.



1.1. Inputs and fixed search

The source curves are the P29 and P37 floor curves named in the ECTD study. The prospective design chose 2,048 P29 and 256 P37 directed edges before enumeration. At every step, the producer required two rational degree-7 neighbors, selected the nonbacktracking neighbor, and stopped on repeated `j`. It transported a generator and a smoke target through each map and checked the dual identity at every edge. The code and raw JSONL are in the linked study directories; the continuation used the same producer on a new write-once output path.

1.2. Observations and independent checks

Both full paths ended with a completion record. The P29 path has 2,048 edges and 2,049 distinct `j` values; P37 has 256 edges and 257 distinct `j` values. The independent checker parsed every edge, checked continuity and unique `j`, recomputed each curve's `j`, proved that the saved cubic kernel divides the seventh division polynomial, recomputed the raw Vélu codomain, and verified the saved generator and smoke target with separate point arithmetic. Both full verification reports passed. The checker does not rederive the complete isogeny point formula; the Sage producer's map and dual checks are a second, implementation-dependent layer.

An earlier P29 attempt stopped after 569 saved edges when the chat session was interrupted. Its prefix and raw file remain unchanged, and the same independent checks pass on those 569 edges. It is an interrupted attempt, not a mathematical counterexample or a yield result.

1.3. Decision boundary

These finite graph results make 2,306 curve presentations available for the prespecified base and target experiment. They do not rank any curve by relation yield or ECDLP cost. The next stage requires a frozen runnable collector, seed commitments, the full base/target controls, and Coordinator approval. A verified improvement compares complete cost `S` against the best frozen index-calculus pipeline for the same curve and size; matched rho and the setup-excluded Bernstein–Lange reference remain reported distances. The protocol amendment is in `AMENDMENT.md`.

1.4. Evidence files

`walk-p29-2048-attempt2.jsonl`, `walk-p37-256.jsonl`, `verification-p29-full.json`, `verification-p37-full.json`, `verify_walk.py`, and `manifest-v2.json` bind this report. The producer, planning draft and interrupted prefix are copied into this continuation so the committed evidence remains self-contained. The first manifest is retained as a superseded draft because it pointed to uncommitted adjacent files.